CLSI 프로그램 소개

연세의료원

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박정용

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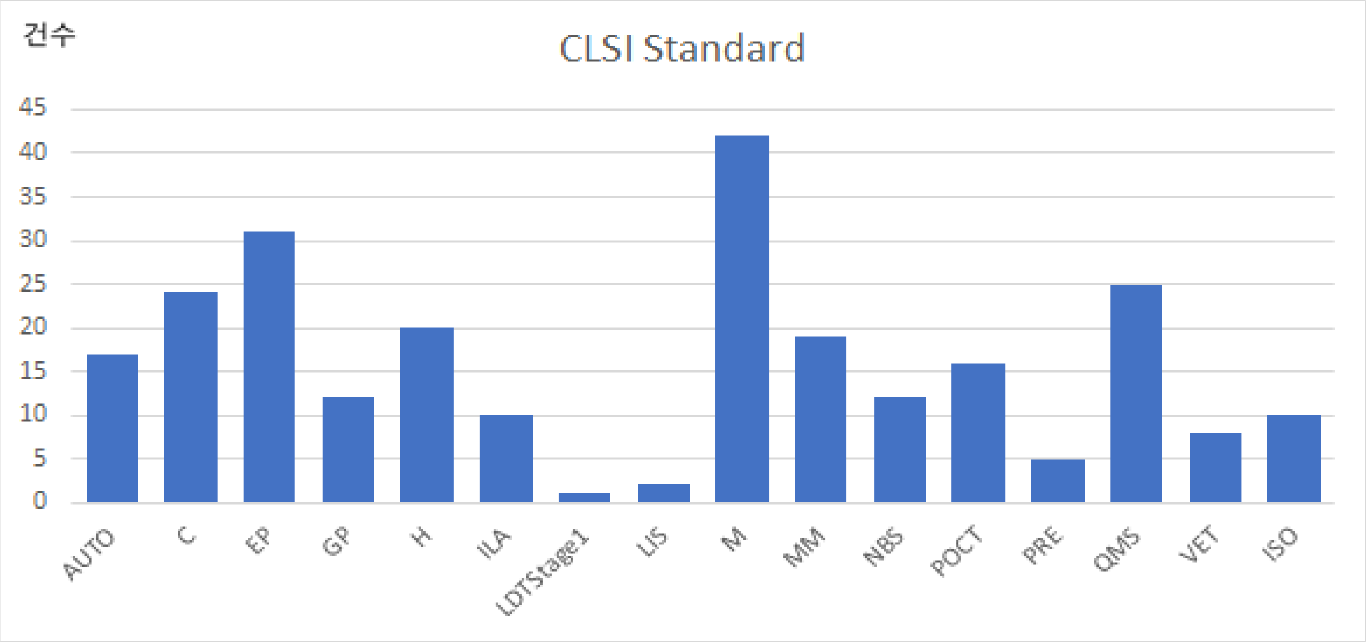


문서 유형	의미	설명
Standard	규범/기준 문서	시험법, 시스템, 프로세스에 대한 표준/요구/권고
Companion	보조자료	Quick Guide, Worksheet, Checklist 등 본문서의 현장 적용 지원
Implementation Guide	이행 가이드	사용자 검증을 실제 실행 단계로 쪼개 설명
Quick Guide	축약 안내서	한 장 또는 짧은 형태로 현장 요약
Supplement / Tables	보충판	AST tables, updated breakpoints 등 주기적 업데이트
Report / White Paper	해설/백서	근거와 사례, 정책적/방법론적 해설
Plus edition	통합판/확장판	최근 CLSI에서 사용하는 최신 판형 중 하나로 본문서+부가자료를 함께 포함하는 경우가 있음

분야
AST(항균제 감수성 시험)
Automation and Informatics(자동화와 정보학)
Clinical Chemistry and Toxicology(임상화학과 독성학)
Hematology(혈액학)
Immunology and Ligand assay(면역학과 리간드분석)
Method Evaluation(검사법 평가)
Microbiology(미생물학)
Molecular Diagnostics(분자진단학)
Newborn Screening(신생아 선별검사)
Point-of-Care Testing(현장검사)
Preexamination Processes(분석전 과정)
Quality Management Systems(품질관리시스템)
Veterinary Medicine(수의학)
LDT(검사실 개발검사)

- **검사 분야별 문서**: 미생물, 혈액, 임상화학/독성, 면역, 분자진단, 신생아, 수의학
- **검사 운영/품질 문서**: 품질관리, 검사 전 과정, 현장검사
- **검사 성능/검증 문서**: Method Evaluation, LDT
- **시스템/기술 문서**: 자동화 및 정보학
- **특수 핵심 영역**: AST

https://clsi.org/shop/standards/

Automation and Informatics

유형	프로그램	제목
Standard	CLSI AUTO01	Laboratory Automation: Specimen Container/Specimen Carrier
Standard	CLSI AUTO02	Laboratory Automation: Bar Codes for Specimen Container Identification
Standard	CLSI AUTO03	Laboratory Automation: Communications With Automated Clinical Laboratory Systems, Instruments, Devices, and Information Systems
Standard	CLSI AUTO04	Laboratory Automation: Systems Operational Requirements, Characteristics, and Information Elements
Standard	CLSI AUTO05	Laboratory Automation: Electromechanical Interfaces
Standard	CLSI AUTO07	Laboratory Automation: Data Content for Specimen Identification
Standard	CLSI AUTO08	Managing and Validating Laboratory Information Systems
Standard	CLSI AUTO09	Remote Access to Clinical Laboratory Diagnostic Devices via the Internet
Standard	CLSI AUTO10	Autoverification of Clinical Laboratory Test Results
Standard	CLSI AUTO11	Information Technology Security of In Vitro Diagnostic Instruments and Software Systems
Standard	CLSI AUTO12	Specimen Labels: Content and Location, Fonts, and Label Orientation
Standard	CLSI AUTO13	Laboratory Instruments and Data Management Systems: Design of Software User Interfaces and End-User Software Systems Validation, Operation, and Monitoring
Standard	CLSI AUTO15	Autoverification of Medical Laboratory Results for Specific Disciplines
Standard	CLSI AUTO16	Next-Generation In Vitro Diagnostic Instrument Interface
Standard	CLSI AUTO17	Semantic Interoperability for In Vitro Diagnostic Systems
Standard	CLSI LIS01	Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems
Standard	CLSI LIS02	Specification for Transferring Information Between Clinical Laboratory Instruments and Information Systems

Method Evaluation

유형	프로그램	제목
Standard	CLSI EP05 Plus	Evaluation of Precision of Quantitative Measurement Procedures
Standard	CLSI EP06	Evaluation of Linearity of Quantitative Measurement Procedures
Companion	CLSI EP06EG	Developer Validation of Linearity Establishment Guide
Companion	CLSI EP06IG	User Verification of Linearity Implementation Guide
Standard	CLSI EP07 Plus	Interference Testing in Clinical Chemistry
Standard	CLSI EP09	Measurement Procedure Comparison and Bias Estimation Using Patient Samples
Standard	CLSI EP10	Preliminary Evaluation of Quantitative Medical Laboratory Measurement Procedures
Companion	CLSI EP10IG	Preliminary Evaluation of Quantitative Medical Laboratory Measurement Procedures Implementation Guide
Standard	CLSI EP12	Evaluation of Qualitative, Binary Output Examination Performance
Companion	CLSI EP12IG	Verification of Performance of a Qualitative, Binary Output Examination Implementation Guide
Companion	CLSI EP14 Guide	Evaluation of Commutability of Processed Samples Guide
Standard	CLSI EP14 Plus	Evaluation of Commutability of Processed Samples
Standard	CLSI EP15	User Verification of Precision and Estimation of Bias
Companion	CLSI EP15IG	User Verification of Precision Implementation Guide
Companion	CLSI EP15IG2	User Verification of Bias (Trueness) Implementation Guide

Method Evaluation

유형	프로그램	제목
Standard	CLSI EP17	Evaluation of Detection Capability for Clinical Laboratory Measurement Procedures
Companion	CLSI EP17IG	Evaluation of Detection Capability Implementation Guide
Standard	CLSI EP18	Risk Management Techniques to Identify and Control Laboratory Error Sources
Companion	CLSI EP18EP23WS	Sources of Failure Template
Companion	CLSI EP18IG	Risk Management Techniques to Identify and Control Laboratory Error Sources Implementation Guide
Standard	CLSI EP19	A Framework for Using CLSI Documents to Evaluate Medical Laboratory Test Methods
Standard	CLSI EP21 Plus	Evaluation of Total Analytical Error for Quantitative Medical Laboratory Measurement Procedures
Standard	CLSI EP23	Laboratory Quality Control Based on Risk Management
Companion	CLSI EP23QG	Developing the Quality Control Plan
Companion	CLSI EP23WB	Laboratory Quality Control Based on Risk Management; Electronic Workbook
Standard	CLSI EP24	Assessment of the Diagnostic Accuracy of Laboratory Tests Using Receiver Operating Characteristic Curves
Standard	CLSI EP25	Evaluation of Stability of In Vitro Medical Laboratory Test Reagents
Standard	CLSI EP26	User Evaluation of Acceptability of a Reagent Lot Change
Standard	CLSI LDTStage1	LDT Regulatory Guidance
Companion	CLSI EPLDTQG	Validating Performance Claims for Laboratory-Developed Tests

Method Evaluation

유형	프로그램	제목
Companion	CLSI EP26IG	User Evaluation of Acceptability of a Reagent Lot Change Implementation Guide
Standard	CLSI EP27	Constructing and Interpreting an Error Grid for Quantitative Measurement Procedures
Standard	CLSI EP28	Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory
Companion	CLSI EP28IG	Verification of Reference Intervals in the Medical Laboratory Implementation Guide
Standard	CLSI EP29	Expression of Measurement Uncertainty in Laboratory Medicine
Standard	CLSI EP30	Characterization and Qualification of Commutable Reference Materials for Laboratory Medicine
Standard	CLSI EP31 Plus	Verification of Comparability of Patient Results Within One Health Care System
Standard	CLSI EP32	Implementation of Metrological Traceability in Laboratory Medicine
Standard	CLSI EP33	Use of Delta Checks in the Medical Laboratory
Standard	CLSI EP34	Establishing and Verifying an Extended Measuring Interval Through Specimen Dilution and Spiking
Companion	CLSI EP34IG	Verification of an Extended Measuring Interval Implementation Guide
Standard	CLSI EP35	Assessment of Equivalence or Suitability of Specimen Types for Medical Laboratory Measurement Procedures
Standard	CLSI EP36	Harmonization of Symbology and Equations
Subscription	CLSI EP37	Supplemental Tables for Interference Testing in Clinical Chemistry
Standard	CLSI EP39 Plus	A Hierarchical Approach to Selecting Surrogate Samples for the Evaluation of In Vitro Medical L aboratory Tests
Standard	CLSI EP43	Implementing a Laboratory Test Under Emergency Use Conditions
Standard	CLSI EP46 Plus	Determining **Allowable Total Error Goals** and Limits for Quantitative Medical Laboratory Measu rement Procedures
Standard	CLSI EP47	Evaluation of Reagent **Carryover** Effects on Test Results
Standard	CLSI EP49	Framework for Developing Evidence of Clinical Validity of Medical Laboratory Test Methods

Immunology & Ligand assay

유형	프로그램	제목
Standard	CLSI ILA02	Quality Assurance of Laboratory Tests for Autoantibodies to Nuclear Antigens: (1) Indirect Fluorescence Assay for Microscopy and (2) Microtiter Enzyme Immunoassay Methods
Standard	CLSI ILA20	Analytical Performance Characteristics, Quality Assurance, and Clinical Utility of Immunological Assays for Human Immunoglobulin E Antibodies of Defined Allergen Specificities
Standard	CLSI ILA21	Clinical Evaluation of Immunoassays
Standard	CLSI ILA23	Assessing the Quality of Immunoassay Systems: Radioimmunoassays and Enzyme, Fluorescence, and Luminescence Immunoassays
Standard	CLSI ILA25	Maternal Serum Screening
Standard	CLSI ILA26	Performance of Single Cell Immune Response Assays
Standard	CLSI ILA28	Quality Assurance for Design Control and Implementation of Immunohistochemistry Assays
Companion	CLSI ILA28QG	Comparison of the Characteristics of Immunoassays Such as Enzyme-Linked Immunosorbent Assay and Immunohistochemistry Quick Guide
Standard	CLSI ILA30	Immunoassay Interference by Endogenous Antibodies
Standard	CLSI ILA33	Validation of Automated Systems for Immunohematological Testing Before Implementation
Standard	CLSI ILA34	Design and Validation of Immunoassays for Assessment of Human Allergenicity of New Biotherapeutic Drugs

Clinical Chemistry and Toxicology

유형	프로그램	제목
Standard	CLSI C24	Statistical Quality Control for Quantitative Measurement Procedures: Principles and Definitions
Standard	CLSI C29	Standardization of Sodium and Potassium Ion-Selective Electrode Systems to the Flame Photometric Reference Method
Standard	CLSI C31	Ionized Calcium Determinations: Precollection Variables, Specimen Choice, Collection, and Handling
Standard	CLSI C34	Sweat Testing: Sample Collection and Quantitative Chloride Analysis
Standard	CLSI C37	Preparation and Validation of Commutable Frozen Human Serum Pools as Secondary Reference Materials for Cholesterol Measurement Procedures
Standard	CLSI C38	Control of Preexamination Variation in Trace Element Determinations
Standard	CLSI C39	A Designated Comparison Method for the Measurement of Ionized Calcium in Serum
Standard	CLSI C40	Measurement Procedures for the Determination of Lead in Whole Blood
Standard	CLSI C42	Erythrocyte Protoporphyrin Testing
Standard	CLSI C43	Gas Chromatography/Mass Spectrometry Confirmation of Drugs
Standard	CLSI C45	Measurement of Free Thyroid Hormones

Clinical Chemistry and Toxicology

유형	프로그램	제목
Standard	CLSI C46	Blood Gas and pH Analysis and Related Measurements
Standard	CLSI C48	Application of Biochemical Markers of Bone Turnover in the Assessment and Monitoring of Bone Diseases
Standard	CLSI C49	Analysis of Body Fluids in Clinical Chemistry
Standard	CLSI C50	Mass Spectrometry in the Clinical Laboratory: General Principles and Guidance
Standard	CLSI C52	Toxicology and Drug Testing in the Medical Laboratory
Standard	CLSI C56	Hemolysis, Icterus, and Lipemia/Turbidity Indices as Indicators of Interference in Clinical Laboratory Analysis
Companion	CLSI C56QG	Examples of Hemolyzed, Icteric, and Lipemic/Turbid Samples Quick Guide
Standard	CLSI C57	Mass Spectrometry for Androgen and Estrogen Measurements in Serum
Standard	CLSI C58	Assessment of Fetal Lung Maturity by the Lamellar Body Count
Standard	CLSI C61	Determination of Serum Iron, Total Iron-Binding Capacity and Percent Transferrin Saturation
Standard	CLSI C62	Liquid Chromatography-Mass Spectrometry Methods
Standard	CLSI C63	Laboratory Support for Pain Management Programs
Standard	CLSI C64	Quantitative Measurement of Proteins and Peptides by Mass Spectrometry
Standard	CLSI C65	Biochemical Tumor Marker Testing

Hematology

유형	프로그램	제목
Standard	CLSI H02	Procedures for the Erythrocyte Sedimentation Rate Test
Standard	CLSI H07	Procedure for Determining Packed Cell Volume by the Microhematocrit Method
Standard	CLSI H15	Reference and Selected Procedures for the Quantitative Determination of Hemoglobin in Blood
Standard	CLSI H20	Reference Leukocyte Differential Count (Proportional) and Evaluation of Instrumental Methods
Standard	CLSI H21	Collection, Transport, and Processing of Blood Specimens for Testing Plasma-Based Coagulation Assays
Companion	CLSI H21QG	Quick Guide for Collection, Transport, and Processing of Blood Specimens for Testing Plasma-Based Coagulation Assays and Molecular Hemostasis Assays
Standard	CLSI H26	Validation, Verification, and Quality Assurance of Automated Hematology Analyzers
Standard	CLSI H30	Procedure for the Determination of Fibrinogen in Plasma
Standard	CLSI H42	Enumeration of Immunologically Defined Cell Populations by Flow Cytometry
Standard	CLSI H43	Clinical Flow Cytometric Analysis of Neoplastic Hematolymphoid Cells
Standard	CLSI H44	Methods for Reticulocyte Counting (Automated Blood Cell Counters, Flow Cytometry, and Supravital Dyes)
Standard	CLSI H47	One-Stage Prothrombin Time (PT) Test and Activated Partial Thromboplastin Time (APTT) Test
Standard	CLSI H48	Determination of Coagulation Factor Activities Using the One-Stage Clotting Assay
Standard	CLSI H52	Red Blood Cell Diagnostic Testing Using Flow Cytometry
Standard	CLSI H54	Procedures for Validation of INR and Local Calibration of PT/INR Systems
Standard	CLSI H56	Body Fluid Analysis for Cellular Composition
Standard	CLSI H57	Protocol for the Evaluation, Validation, and Implementation of Coagulometers
Standard	CLSI H58	Platelet Function Testing by Aggregometry
Standard	CLSI H59	Quantitative D-dimer for the Exclusion of Venous Thromboembolic Disease
Standard	CLSI H60	Laboratory Testing for the Lupus Anticoagulant
Companion	CLSI H60QG1	Algorithmic Approach to Lupus Anticoagulant Testing
Companion	CLSI H60QG2	Criteria for the Laboratory Diagnosis of the Lupus Anticoagulant
Standard	CLSI H62	Validation of Assays Performed by Flow Cytometry

Quality Management System

유형	프로그램	제목
Standard	CLSI QMS01	A Quality Management System Model for Laboratory Services
Companion	CLSI QMS01CL	Gap Analysis Checklists, 1st Edition
Standard	CLSI QMS01es	A Quality Management System Model for Laboratory Services
Standard	CLSI QMS02	Developing and Managing Laboratory Documents
Companion	CLSI QMS02QG	Ten Rules for Laboratory Document Management Quick Guide, 7th Edition
Standard	CLSI QMS03	Training and Competence Assessment
Standard	CLSI QMS04	Laboratory Design
Standard	CLSI QMS05	Qualifying, Selecting, and Evaluating a Referral Laboratory
Standard	CLSI QMS06	Quality Management System: Continual Improvement
Standard	CLSI QMS11	Nonconforming Event Management
Standard	CLSI QMS12	Developing and Using Quality Indicators for Laboratory Improvement
Standard	CLSI QMS13A	Quality Management System: Equipment
Standard	CLSI QMS14	Quality Management System: Leadership and Management Roles and Responsibiliti es
Standard	CLSI QMS15	Laboratory Internal Audit Program
Standards Package	CLSI LQMS PKG	Laboratory Quality Management System Certificate Program Package
Standards Package	CLSI LQMS PKG ES	Paquete digital de Programa de Certificación del CLSI en el Sistema de Gestión de l a Calidad en el Laboratorio

Quality Management System

유형	프로그램	제목
Standard	CLSI QMS16	Laboratory Personnel Management
Standard	CLSI QMS17	External Assessments, Audits, and Inspections of the Laboratory
Standard	CLSI QMS18	Process Management
Standard	CLSI QMS19	Customer Focus in a Quality Management System
Standard	CLSI QMS20	The Cost of Quality in Medical Laboratories
Standard	CLSI QMS21	Purchasing and Inventory Management
Standard	CLSI QMS22	Management of Paper-based and Electronic Laboratory Information
Standard	CLSI QMS23	General Laboratory Equipment Performance Qualification, Use, and Maintenance
Standard	CLSI QMS24	Using Proficiency Testing and Alternative Assessment to Improve Medical Laboratory Quality
Standard	CLSI QMS25	Handbook for Developing a Laboratory Quality Manual
Standard	CLSI QMS26	Managing Laboratory Records
Standard	CLSI QMS28	Laboratory Safety Management
Standard	CLSI QMS29	Management Review

Microbiology

유형	프로그램	제목
Standard	CLSI M02	Performance Standards for Antimicrobial Disk Susceptibility Tests
Companion	CLSI M02QG	Disk Diffusion Reading Guide
Standard	CLSI M07	Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically
Companion	CLSI M07QG	M07Ed12QGE Minimal Inhibitory Concentration Reading Guide
Standard	CLSI M100	Performance Standards for Antimicrobial Susceptibility Testing
Standard	CLSI M100JA	Performance Standards for Antimicrobial Susceptibility Testing Japanese Edition
Subscription	CLSI M100Plus	M100 Plus
Standard	CLSI M11	Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria
Standard	CLSI M15	Laboratory Diagnosis of Blood-borne Parasitic Diseases
Standard	CLSI M22	Quality Control for Commercially Prepared Microbiological Culture Media; Approved Standard
Standard	CLSI M23	Development of In Vitro Susceptibility Test Methods, Breakpoints, and Quality Control Parameters
Standard	CLSI M23S	Procedure for Optimizing Disk Contents (Potencies) for Disk Diffusion Testing of Antimicrobial Age nts Using Harmonized CLSI and EUCAST Criteria
Standard	CLSI M23S2	Process to Submit Disk Content (Potency) Data for Joint CLSI-EUCAST Working Group Review and Approval
Standard	CLSI M23S3	Procedure for Confirming the Acceptability of Mueller-Hinton Agar Sources for Subsequent Use in CLSI and/or EUCAST Studies to Establish Disk Diffusion Quality Control Ranges
Standard	CLSI M24	Susceptibility Testing of Mycobacteria, Nocardia spp., and Other Aerobic Actinomycetes
Standard	CLSI M24S	Performance Standards for Susceptibility Testing of Mycobacteria, Nocardia spp., and Other Aerobi c Actinomycetes
Standard	CLSI M26	Methods for Determining Bactericidal Activity of Antimicrobial Agents
Standard	CLSI M27	Reference Method for Broth Dilution Antifungal Susceptibility Testing of Yeasts
Standard	CLSI M27M44S	Performance Standards for Antifungal Susceptibility Testing of Yeasts
Standard	CLSI M28	Procedures for the Recovery and Identification of Parasites From the Intestinal Tract
Standard	CLSI M29	Protection of Laboratory Workers From Occupationally Acquired Infections

Microbiology

유형	프로그램	제목
Standard	CLSI M34	Western Blot Assay for Antibodies to Borrelia burgdorferi
Standard	CLSI M35	Abbreviated Identification of Bacteria and Yeast
Standard	CLSI M36	Clinical Use and Interpretation of Serologic Tests for Toxoplasma gondii
Standard	CLSI M38	Reference Method for Broth Dilution Antifungal Susceptibility Testing of Filamentous Fungi
Standard	CLSI M38M51S	Performance Standards for Antifungal Susceptibility Testing of Filamentous Fungi
Standard	CLSI M39	Analysis and Presentation of Cumulative Antimicrobial Susceptibility Test Data
Standard	CLSI M40	Quality Control of Microbiological Transport Systems; Approved Standard
Standard	CLSI M41	Viral Culture
Standard	CLSI M43	Methods for Antimicrobial Susceptibility Testing for Human Mycoplasmas
Standard	CLSI M44	Method for Antifungal Disk Diffusion Susceptibility Testing of Yeasts
Standard	CLSI M45	Methods for Antimicrobial Dilution and Disk Susceptibility Testing of Infrequently Isolated or Fastidious Bacteria
Standard	CLSI M47	Principles and Procedures for Blood Cultures
Standard	CLSI M48	Laboratory Detection and Identification of Mycobacteria
Standard	CLSI M50	Quality Control for Commercial Microbial Identification Systems; Approved Guideline
Standard	CLSI M51	Method for Antifungal Disk Diffusion Susceptibility Testing of Nondermatophyte Filamentous Fungi
Standard	CLSI M52	Verification of Commercial Microbial Identification and Antimicrobial Susceptibility Testing Systems
Standard	CLSI M53	Criteria for Laboratory Testing and Diagnosis of Human Immunodeficiency Virus Infection
Standard	CLSI M54	Principles and Procedures for Detection and Culture of Fungi in Clinical Specimens
Standard	CLSI M56	Principles and Procedures for Detection of Anaerobes in Clinical Specimens
Standard	CLSI M57	Principles and Procedures for the Development of Epidemiological Cutoff Values for Antifungal Susceptibility Testing

Microbiology

유형	프로그램	제목
Standard	CLSI M57S	Epidemiological Cutoff Values for Antifungal Susceptibility Testing
Standard	CLSI M58	Methods for the Identification of Cultured Microorganisms Using Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry
Standard	CLSI M64	Implementation of Taxonomy Nomenclature Changes
Standard	CLSI M67	Verification of Laboratory Automation in Microbiology
Companion	CLSI MR01	Polymyxin Breakpoints for Enterobacterales, Pseudomonas aeruginosa, and Acinetobacter spp.
Companion	CLSI MR02	Fluoroquinolone Breakpoints for Enterobacteriaceae and Pseudomonas aeruginosa
Companion	CLSI MR03	Meropenem Breakpoints for Acinetobacter spp.
Companion	CLSI MR04	Azithromycin Breakpoint for Neisseria gonorrhoeae
Companion	CLSI MR05	Ceftaroline Breakpoints for Staphylococcus aureus
Companion	CLSI MR06	Daptomycin Breakpoints for Enterococci
Companion	CLSI MR07	Cefazolin Breakpoints for Enterobacterales (Systemic Infections)
Companion	CLSI MR08	Cefazolin Breakpoints for Enterobacterales (Uncomplicated Urinary Tract Infections)
Companion	CLSI MR14	Piperacillin-Tazobactam Breakpoints for Enterobacterales
Companion	CLSI MR15	Piperacillin-Tazobactam Breakpoints for Pseudomonas aeruginosa
Companion	CLSI MR16	Aminoglycoside Breakpoints for Enterobacterales and Pseudomonas aeruginosa

Microbiology-AST

유형	프로그램	제목
Standards Package	CLSI AST M02 PKG	M02-Ed14 and M100-Ed36PK26 Package of 2 Documents
Standards Package	CLSI AST M02 PKG Print	M02-Ed14 and M100-Ed36PK26 Package of 2 Documents
Standards Package	CLSI AST M07 PKG	M07-Ed12 and M100-Ed36 Package
Standards Package	CLSI AST M07 PKG Print	M07-Ed12 and M100-Ed36 Package of 2 Documents - Print
Standards Package	CLSI AST M100 PKG	M02-Ed14 M07-Ed12 and M100Ed36 Package of 3 documents
Standards Package	CLSI AST M100 PKG Print	M02-Ed14 M07-Ed12 and M100Ed36 Package of 3 documents - Print
Standards Package	CLSI AST Rationale PKG	AST Rationale Documents (Free)
Companion	CLSI FR01	Voriconazole Breakpoints for Aspergillus fumigatus
Companion	CLSI FR02	Isavuconazole Breakpoints for Aspergillus fumigatus

Molecular Diagnostics

유형	프로그램	제목
Standard	CLSI MM01	Molecular Testing for Heritable Genetics and Specimen Identification
Standard	CLSI MM03	Molecular Diagnostic Methods for Infectious Diseases
Standard	CLSI MM05	Nucleic Acid Amplification Assays for Molecular Hematopathology
Standard	CLSI MM06	Quantitative Molecular Methods for Infectious Diseases
Standard	CLSI MM07	Fluorescence In Situ Hybridization Methods for Clinical Laboratories; Approved Guideline
Standard	CLSI MM09	Human Genetic and Genomic Testing Using Traditional and High-Throughput Nucleic Acid Sequencing Methods
Standard	CLSI MM11	Molecular Methods for Bacterial Strain Typing
Standard	CLSI MM12	Diagnostic Nucleic Acid Microarrays
Standard	CLSI MM13	Collection, Transport, Preparation, and Storage of Specimens for Molecular Methods
Standard	CLSI MM14	Design of Molecular Proficiency Testing/ External Quality Assessment
Standard	CLSI MM17	Validation and Verification of Multiplex Nucleic Acid Assays
Standard	CLSI MM18	Interpretive Criteria for Identification of Bacteria and Fungi by Targeted DNA Sequencing
Standard	CLSI MM19	Establishing Molecular Testing in Medical Laboratory Environments
Standard	CLSI MM20	Quality Management for Molecular Genetic Testing; Approved Guideline
Standard	CLSI MM21	Genomic Copy Number Microarrays for Constitutional Genetic and Oncology Applications
Standard	CLSI MM22	Microarrays for Diagnosis and Monitoring of Infectious Diseases
Standard	CLSI MM23	Molecular Diagnostic Methods for Solid Tumors (Nonhematological Neoplasms)
Standard	CLSI MM24	Molecular Methods for Genotyping and Strain Typing of Infectious Organisms
Standard	CLSI MM26	Cancer Molecular Testing: Principles of Oncology Test Interpretation, Laboratory and Assay Design, and Clinical Consultation

Point-of-Care Testing

유형	프로그램	제목
Standard	CLSI POCT01	Point-of-Care Connectivity
Companion	CLSI POCT01QG	CLSI Vendor Code Registry
Standard	CLSI POCT02	Implementation Guide of POCT01 for Health Care Providers
Standard	CLSI POCT04	Essential Tools for Implementation and Management of a Point-of-Care Testing Program
Standard	CLSI POCT05	Performance Metrics for Continuous Interstitial Glucose Monitoring
Standard	CLSI POCT06	Effects of Different Sample Types on Glucose Measurements
Standard	CLSI POCT07	Quality Management: Approaches to Reducing Errors at the Point of Care
Standard	CLSI POCT08	Quality Practices in Noninstrumented Point-of-Care Testing: An Instructional Manual and Resources for Health Care Workers
Companion	CLSI POCT08QG1	Corrective Action Quick Guide
Companion	CLSI POCT08QG2	Quality Control Troubleshooting Flow Chart
Companion	CLSI POCT08QG3	Quality Control Log Sheet Quick Guide
Standard	CLSI POCT09	Selection Criteria for Point-of-Care Testing Devices
Companion	CLSI POCT09WS	Instrument Selection Worksheet
Standard	CLSI POCT10	Physician and Nonphysician Provider-Performed Microscopy Testing
Companion	CLSI POCT10 Checklist 1	Provider-Performed Microscopy Competency Assessment
Companion	CLSI POCT10 Checklist 2	Provider-Performed Microscopy Training Checklist
Companion	CLSI POCT10 Quick Guide	Provider-Performed Microscopy Training Quick Guide

Point-of-Care Testing

유형	프로그램	제목
Standard	CLSI POCT12	Point-of-Care Blood Glucose Testing in Acute and Chronic Care Facilities; Approved Guideline
Standard	CLSI POCT13	Glucose Monitoring in Settings Without Laboratory Support
Standard	CLSI POCT14	Point-of-Care Coagulation Testing and Anticoagulation Monitoring
Standard	CLSI POCT15	Point-of-Care Testing for Infectious Diseases
Standard	CLSI POCT16	Emergency and Disaster Point-of-Care Testing
Standard	CLSI POCT17	Use of Glucose Meters for Critically Ill Patients
Standard	CLSI POCT18	Selection Process for CLIA-Waived Testing for SARS-CoV-2, Respiratory Syncytial Virus, and Influenza Viruses

Veterinary Medicine

유형	프로그램	제목
Standard	CLSI VET01	Performance Standards for Antimicrobial Disk and Dilution Susceptibility Tests for Bacteria Isolated From Animals
Standard	CLSI VET01S	Performance Standards for Antimicrobial Disk and Dilution Susceptibility Tests for Bacteria Isolated From Animals
Standard	CLSI VET02	Development of Quality Control Ranges, Breakpoints, and Interpretive Categories for Antimicrobial Agents Used in Veterinary Medicine
Standard	CLSI VET03	Methods for Antimicrobial Broth Dilution and Disk Diffusion Susceptibility Testing of Bacteria Isolated From Aquatic Animals
Standard	CLSI VET04	Performance Standards for Antimicrobial Susceptibility Testing of Bacteria Isolated From Aquatic Animals
Standard	CLSI VET05	Generation, Presentation, and Application of Antimicrobial Susceptibility Test Data for Bacteria of Animal Origin
Standard	CLSI VET06	Methods for Antimicrobial Susceptibility Testing of Infrequently Isolated or Fastidious Bacteria Isolated From Animals
Standard	CLSI VET09	Understanding Susceptibility Test Data as a Component of Antimicrobial Stewardship in Veterinary Settings

Newborn Screening

유형	프로그램	제목
Standard	CLSI NBS01	Dried Blood Spot Specimen Collection for Newborn Screening
Companion	CLSI NBS01QG	Specimen Collection and Specimen Quality for Newborn Screening Quick Guide
Standard	CLSI NBS02	Newborn Screening Follow-up and Education
Standard	CLSI NBS03	Newborn Screening for Preterm, Low Birth Weight, and Sick Newborns
Standard	CLSI NBS04	Newborn Screening by Tandem Mass Spectrometry
Standard	CLSI NBS05	Newborn Screening for Cystic Fibrosis
Standard	CLSI NBS06	Newborn Screening for Severe Combined Immunodeficiency and Other Related Severe T-cell and B-cell Immunodeficiencies
Standard	CLSI NBS07	Newborn Blood Spot Screening for Pompe Disease by Lysosomal Acid a-Glucosidase Activity Assays
Standard	CLSI NBS08	Newborn Screening for Hemoglobinopathies
Standard	CLSI NBS09	Newborn Screening for X-Linked Adrenoleukodystrophy
Standard	CLSI NBS10	Newborn Screening for Congenital Hypothyroidism
Standard	CLSI NBS11	Newborn Screening for Congenital Adrenal Hyperplasia
Standard	CLSI NBS13	Newborn Screening for Spinal Muscular Atrophy

ISO

유형	프로그램	제목
Standard	ISO/TS 22583	Requirements and Recommendations for Supervisors and Operators of Point-of-Care Testing (POCT) Equipment, 2nd Edition
Standard	ISO/TS2 3824	Medical laboratories — Guidance on application of ISO 15189 in anatomic pathology
Standard	ISO15189	Medical laboratories - Requirements for quality and competence
Standard	ISO15190	Medical laboratories — Requirements for safety, 2nd Edition
Standard	ISO17511	In vitro diagnostic medical devices — Requirements for establishing metrological traceability of values assigned to calibrators, trueness control materials and human samples, 2nd Edition
Standard	ISO20916	In vitro diagnostic medical devices -- Clinical performance studies using specimens from human subjects -- Good study practice
Standard	ISO21474-1	In vitro diagnostic medical devices — Multiplex molecular testing for nucleic acids — Part 1: Terminology and general requirements for nucleic acid quality evaluation
Standard	ISO21474-2	In vitro diagnostic medical devices — Multiplex molecular testing for nucleic acids — Part 2: Validation and verification
Standard	ISO21474-3	In vitro diagnostic medical devices — Multiplex molecular testing for nucleic acids Part 3: Interpretation and reports
Standard	ISO23640	In vitro diagnostic medical devices — Evaluation of stability of in vitro diagnostic reagents

Preexamination Processes

유형	프로그램	제목
Standard	CLSI PRE01	Patient and Laboratory Specimen Identification Processes
Standard	CLSI PRE02	Collection of Diagnostic Venous Blood Specimens
Companion	CLSI PRE02QG	Collection of Diagnostic Venous Blood Specimens Quick Guide
Standard	CLSI PRE04	Handling, Transport, Processing, and Storage of Blood Specimens for Routine Laboratory Examinations
Standard	CLSI PRE05	Processes for the Collection of Urine Specimens
Standard	CLSI PRE06	Evaluation of External Transport Systems

Others

유형	프로그램	제목
Standard	CLSI GP15	Cervicovaginal Cytology Based on the Papanicolaou Technique
Standard	CLSI GP20	Fine Needle Aspiration Biopsy (FNAB) Techniques
Standard	CLSI GP23	Nongynecological Cytology Specimens: Preexamination, Examination, and Postexamination Processes
Standard	CLSI GP34	Validation and Verification of Tubes for Venous and Capillary Blood Specimen Collection
Standard	CLSI GP36	Planning for Laboratory Operations During a Disaster
Standard	CLSI GP39	Tubes and Additives for Venous and Capillary Blood Specimen Collection
Standard	CLSI GP40	Preparations and Testing of Reagent Water in the Medical Laboratory
Standard	CLSI GP42	Collection of Capillary Blood Specimens
Standard	CLSI GP45	Studies to Evaluate Patient Outcomes
Standard	CLSI GP47	Management of Critical- and Significant-Risk Results
Standard	CLSI GP48	Essential Elements of a Phlebotomy Training Program
Standard	CLSI GP49	Developing and Managing a Medical Laboratory (Test) Utilization Management Program

Prefix / 프로그램	공식 분야명	핵심 성격	주로 다루는 내용	주 사용자	대표 문서
EP	Method Evaluation	정량/ 정성 검사법 성능평가	정밀도, 선형성, 비교시험, 검출한계, 참고구간, total error, carryover, reagent lot 평가	생화학, 면역, 분자, 일반 진단검사	EP05, EP06, EP09, EP12, EP15, EP17, EP28, EP47, EP49
C	Clinical Chemistry Toxicology	임상화학/ 독성학	화학·질량분석·바이오마커· 간섭·전처리 변이 관리	생화학, 독성학, 특수화학	C24, C31, C38, C56, C64, C65
H	Hematology	혈액학/응고/유세포	혈구계수기 검증, 응고검사 전처리, 유세포분석, body fluid, lupus anticoagulant	혈액학, 응고검사, 유세포	H21, H26, H42, H56, H57, H60
Immunology / Ligand Assay	Immunology & Ligand Assay	면역학/ 리간드 검사	전용 단일 prefix보다 EP·C·H 문서를 조합해 운영하는 경우가 많음	진단면역, 특수면역	EP12, EP17, C56, H42
M	Microbiology	미생물학	배양, blood culture, transport system, identification, safety, AST 보조문서	미생물실	M22, M29, M39, M40, M41, M45, M47, M52, M56, M67
AST (M series core)	Antimicrobial Susceptibility Testing	AST 핵심 프로그램군	disk diffusion, dilution, breakpoint, QC range, interpretive criteria	미생물실, AST	M02, M07, M11, M23, M100
MM	Molecular Diagnostics	분자진단	유전질환, 감염, 종양, sequencing, proficiency testing, specimen handling	분자진단실	MM01, MM03, MM05, MM09, MM14, MM17, MM19, MM23
QMS	Quality Management	품질경영시스템	문서관리, 교육·역량평가, 감사, 장비, 비일치관리, 정보관리, 프로세스관리	전 검사실	QMS01, QMS02, QMS03, QMS14, QMS18, QMS22
AUTO	Automation and Informatics	자동화/정보학	LIS, autoverification, 보안, 인터페이스, 데이터 교환	전산/자동화 담당, 핵심검사실	AUTO03, AUTO08, AUTO10, AUTO11, AUTO15
NBS	Newborn Screening	신생아선별	dried blood spot, follow-up, 질환별 screening guidance	신생아선별실	NBS01, NBS02, NBS06, NBS08, NBS10, NBS13
POCT	Point-of-Care Testing	현장검사	POCT 프로그램 구축, 기기 선택, 오류감소, 비기기형 검사 품질, CGM	병동 POCT, 응급실, 외래	POCT04, POCT05, POCT07, POCT08, POCT09
PRE	Preexamination Processes	검사 전단계	환자/검체 식별, 채혈, 운송, 보관, 소변 수집, 외부 운송시스템 평가	채혈실, 접수, 전처리	PRE01, PRE02, PRE04, PRE05, PRE06
GP	General / legacy practices	일반실무/ 이행기 문서	과거 GP prefix 문서들. 일부는 PRE/QMS로 재편	채혈, 일반실무	GP34, GP39, GP40, GP42, GP47, GP48
VET	Veterinary Medicine	수의학	동물 유래 균 AST, breakpoint, QC, method	수의진단	VET01, VET01S, VET02
MR	AST rationale/support	AST 근거문서	breakpoint rationale 등 AST 근거 설명 문서	AST 전문가	MR14 등
ISO (distributed by CLSI)	ISO standards via CLSI shop	ISO 연계	ISO 15189 등 시험실·IVD 관련 국제표준을 CLSI shop에서 함께 제공	품질/인증 담당	ISO15189, ISO17593, ISO18113 series, ISO20166 series
LDT support	Laboratory Developed Tests	LDT 지원	새 FDA oversight 대응용 안내, EP49, Method Navigator 등 cross-cutting 자원	개발실, LDT 담당	EP49, Method Navigator

출처: https://clsi.org/standards-development/specialty-areas/

Immunology, Ligand Assay, I/LA Family

Code	문서명	핵심 용도	비고
I/LA02	Quality assurance of laboratory tests for autoantibodies to nuclear antigens: (1)Indirect fluorescence assay for microscopy and (2) Microtiter enzyme immunoassay methods	ANA 검사 품질관리	ANA
I/LA20	Analytical performance characteristics, quality assurance, and clinical utility of immunological assays for human Immunolglobulin E antibodies of defined allergen specificities	Allergy Ig E 검사 문서	IgE
I/LA21	Clinical Evaluation of immunoassays	신규 면역검사 임상평가	EIA
I/LA23	Assessing the quality of immunoassay systems: Radioimmunoassays and enzyme, fluorescence, and luminescence immunoassays	면역측정시스템 자체의 품질평가 문서	RIA, ELISA FIA, CLIA etc.
I/LA25	Maternal serum screening	산전선별검사	
I/LA26	Performance of single cell immune response assays	단일세포 수준의 면역기능평가	
I/LA28	Quality assurance for design control and implementation of immunohistochemistry assays; approved guideline	면역조직화학문서	
I/LA30	Immunoassay interference by endogenous antibodies	endogenous antibodies interference	
I/LA33	Validation of automated systems for immunohematological testing before implementation	immunohematological testing	
I/LA34	Design and validation of immunoassays for assessment of human allergenicity for new biotherapeutic drugs	바이오 의약품 알레르기성 평가 문서	

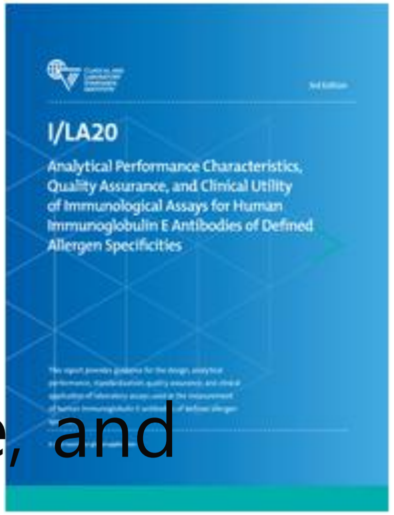
CLSI ILA02

- Quality Assurance of Laboratory Tests for Autoantibodies to Nuclear Antigens:

(1) Indirect Fluorescence Assay for Microscopy and

(2) Microtiter Enzyme Immunoassay Methods

CLSI ILA20



- Analytical Performance Characteristics, Quality Assurance, and Clinical Utility of Immunological Assays for Human Immunoglobulin E Antibodies of Defined Allergen Specificities


Archived Document

I/LA21-A2

Clinical Evaluation of Immunoassays; Approved Guideline—Second Edition

CLSI ILA21

- Clinical Evaluation of Immunoassays
  - Need for clinical evaluation of new immunoassays and new applications of Existing assays, as well as multiple assay Formats and their uses.

    This document will aid developers of "in-house" assays for institutional use

CLSI ILA23

- Assessing the Quality of Immunoassay Systems: Radioimmunoassays and Enzyme, Fluorescence, and Luminescence Immunoassays


Archived Document

I/LA23-A

Assessing the Quality of Immunoassay Systems: Radioimmunoassays and Enzyme, Fluorescence, and Luminescence Immunoassays; Approved Guideline

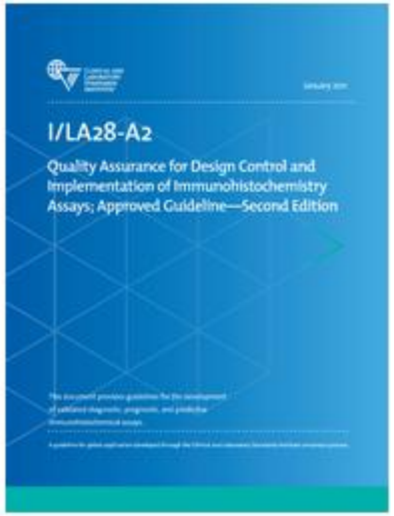
CLSI ILA25

- Maternal Serum Screening

CLSI ILA26

- Performance of Single Cell Immune Response Assays

CLSI ILA28



- Quality Assurance for Design Control and Implementati Immunohistochemistry Assays

CLSI ILA30

- Immunoassay Interference by Endogenous Antibodies

CLSI ILA33

- Validation of Automated Systems for Immunohematological Testing Before Implementation

Contents (Continued)

CLSI ILA34

- Design and Validation of Immunoassays for Assessment of Human Allergenicity of New Biotherapeutic Drugs

11 Assay Interference 29

11.1 Drug-Specific IgG Antibody 30

검사방법 성능의 요소

- 정밀도(precision)
- 정확도(accuracy)
- 직선성(linearity)
- 검출한계, 정량한계(LOD, LOQ)
- 간섭효과(Interference)
- 기질효과(matrix effects)
- 참고범위(reference interval)
- 잔효(carry over)
- 안정성(stability)
- 기타 검사방법별 특이적 요소

면역, 생화학, 혈액학

구분	핵심 CLSI 문서	주 사용 분야	실무에서 쓰는 상황	핵심 포인트
면역학	EP05, EP07, EP09, EP12, EP15, EP17, EP28	면역검사	정량/정성검사 검증, cut-off 검토, 간섭·hook effect 점검	정성·정량 혼재, 판정기준 관리 중요
생화학	EP05, EP06, EP07, EP09, EP15, EP17, EP28	화학/면역 공통	신규 장비 검증, QC 설정, 선형성, 간섭평가, 기준범위	정량검사 중심, TEa/QC 연결이 쉬움
혈액학	H26, H20, EP05, EP09, EP15	CBC/WBC diff/ 특수혈액	자동혈구분석기 검증, 장비 비교, 재현성 평가	세포 분류 특성상 변동성·플래그 해석 중요

- 정밀도란- 동일하거나 유사한 대상에 대해 지정한 조건에서 반복 측정을 통해 얻은 값 또는 측정값간의 일치 정도
- 체외진단의료기기 제조사, 개발자가 검사법을 개발하거나 임상검사실에서 신규 검사법을 도입할 때 성능이 제조사의 주장과 일치하는지 확인하기위해 정밀도평가를 시행
- 다른 성능지표를 평가하기 전에 중요한 요소
- 정밀도? EP05-A3, EP15-A3, EP10-A3
- 20X2X2설계법(단일장소설계), 3X5X5(여러장소(다기관)설계) 등
  - 예비평가? →EP10-A3
- 참값(또는 참고치)과는 무관한 특정 값에 대한 반복 측정의 변동성
- SD, V, CV같은 비정밀도(imprecision)

- 같은 검체를 반복 측정했을 때 결과가 얼마나 일정한가
- 정밀도의 종류
  - 반복정밀도(repeatability)(=within-run precision)--**검사차례내정밀도**
  - 중간정밀도(intermediate precision)(=within device(lab) precision, =total precision)—**검사실내정밀도**=검사장비내 정밀도, 전체정밀도
  - 재현성(reproducibility)(=between-lab, interlab, among-lab precision)--**검사실간정밀도**

반복정밀도, 검사실내비정밀도

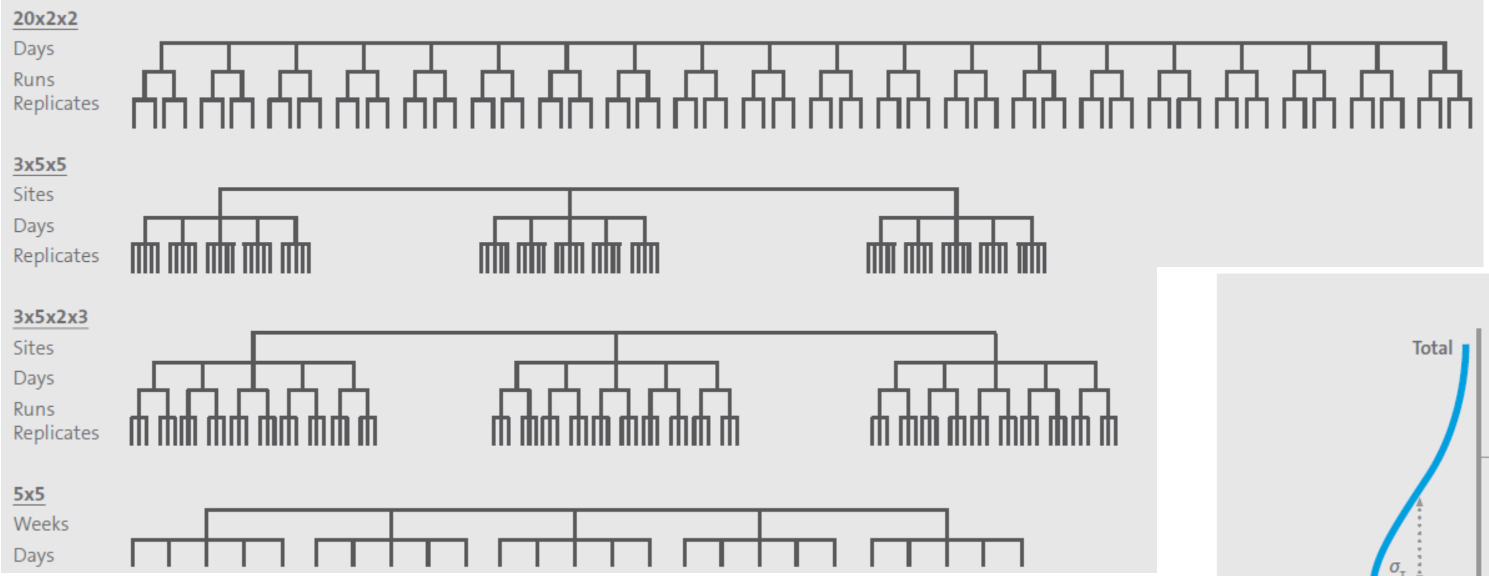


Figure 3. "Comb" Diagrams

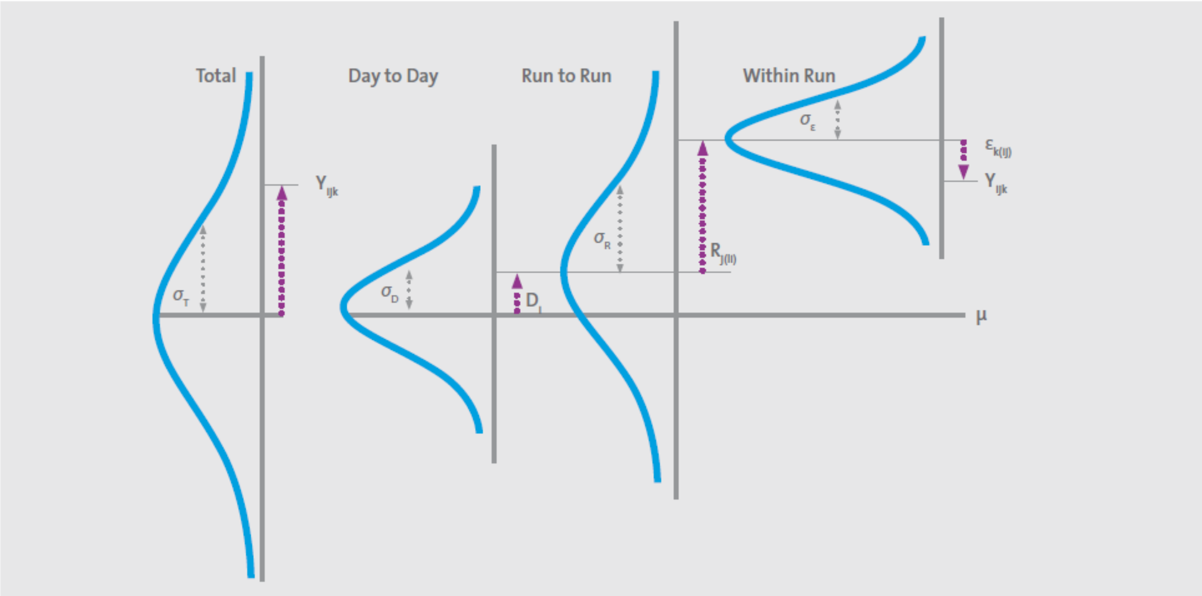


Figure 4. Variance Decomposition. An Individual Result Y_{ijk} (ie, the result for a single replicate) in a 20×2×2 study, modeled as the vector sum of deviations—$D_i + R_{j(i)} + \varepsilon_{k(ij)}$—from the sample's mean, μ, due to the combined effect of three composite sources of random variation—namely, day-to-day (*D*), run-to-run within-day (*R*), and within-run or residual (*ε*) variance components—with distributions characterized by SDs of σ_D, σ_R, and σ_ε, respectively. The heavy, single-headed arrows represent the magnitude and direction of the overall (total) and three source-specific deviations (shifts). The thin, double-headed arrows represent the magnitudes of the respective SDs.

- 정밀도 평가용검체
  - 의학적 의사결정농도를 포함하는 혼합 환자 검체, 정도관리물질 또는 보정물질

- 정밀도에 대한 제조사주장(claim)과 부여된 값에 대한 bias를 동시에 검정(verification)하는 프로토콜
- 대상은 임상 검사실(제조사의 주장에 부합하는지 검정)
- 5X5설계법(최소 두 개이상의 검체를 최소 5일동안 5번의 반복측정을 수행하여 반복정밀도, 검사실내비정밀도 계산, CV평가

  - **EP05-A3** (정량 측정법의 정밀도 평가): 제조사나 개발자가 검사법의 정밀도 성능을 확립하기 위한 표준 지침
  - **EP15-A3:** 검사실 현장에서 제조사가 주장하는 정밀도와 정확도 성능을 5일 이내의 짧은 기간 동안 검증할 때 사용
  - 반복정밀도, 검사실내비정밀도가 각각 제조사에서 제시한 비정밀도 성능보다 유의하게 크지 않음을 검정

- Precision estimation, verification
  - Within-lab precision, repeatability
  - 제조사 CV와 우리 검사실 CV 비교
- Bias estimation, verification
  - Bias=측정값-기준값
  - Bias≤허용오차

- 왜 EP15를 해야 하는가?
  - 제조사 데이터는 우리 환경하의 결과와 다르다(숙련된 실험과 최적의 조건과는 달리 환경, 검사자, 다른 QC 등으로 차이발생)
  - CLIA/ISO15189 요구 (제조사 성능 검증이 필요하다)
  - 환자 안전

  ➔absolute performance확인

  ➔면역학적 검사는?

EP15 User Verification of Precision and Estimation of Bias

Variable	EP05-A3		EP15-A3
Design	20×2×2 design	3×5×5 design	5×5 design
No. of samples	Three levels or five or more levels	Three levels or five or more levels Same samples at each site	At least two samples
Measurand levels	Manufacturer's specified measuring interval, covering points near important medical decision levels		
No. of site	Single site	At least three sites	Single site
No. of instruments	Single instrument	Single instrument	Single instrument
No. of reagent lots	Single reagent lot	Same reagent lot at each site	Single reagent lot
No. of calibrator lots	Single calibrator lot	Same calibrator lot at each site	Single calibrator lot
No. of calibration cycles	Calibration at the start of the study and subsequent calibrations performed per the manufacturer's instruction		
No. of days	20 or more testing days	Five or more days	Five or more days
No. of runs	Two runs per day	Single run	Single run
No. of replicates	Two replicates per run	Five replicates per run	Five replicates per run
Statistics	Two-way nested ANOVA	Two-way nested ANOVA One-way ANOVA	One-way ANOVA
Providing estimates	Repeatability Within-laboratory precision	Repeatability Within-laboratory precision Reproducibility	Repeatability Within-laboratory precision
Purpose	· Establish precision · Re-assess precision for modifications to an existing assay · Understand and assess precision to learn how estimates are established and conduct rigorous precision assessments		· Verify assay precision · Evaluate precision performance as part of a corrective action plan · Assess precision as part of troubleshooting efforts

- 선형성 평가

Standard	CLSI EP06	Evaluation of Linearity of Quantitative Measurement Procedures
Companion	CLSI EP06EG	Developer Validation of Linearity Establishment Guide
Companion	CLSI EP06IG	User Verification of Linearity Implementation Guide

- 어떤 농도구간에서 측정값이 무작위 오차를 제외하면 참값에 비례하여 변해야 함=기대되는 비례관계를 따르느냐?
- LLoQ와 ULoQ를 포함하는 농도구간에서 선형성 claim을 verify 하려는 실험실, validate하려는 제조사,개발자 대상
- 선형성탐구의 2가지 기본 설계
  - 고농도 검체+measurand-free diluent를 이용한 전통적 방법
  - High, low시료의 concentration ratio를 이용한 방법

- Verificaiton
  - 5 point를 찍고, r^2를 본다~~ ➔ 농도별 이질분산을 반영한 1차 회귀와 신뢰구간을 사용하라
  - 1차 회귀를 기본으로 하되 각 농도점에서 허용가능한 deviation을 확인하라
- 임상적으로 의미있는 구간에서 허용가능한 비선형 편차를 관리
- 기본 구조
  - L1(100%)고농도, L2(75%), L3(50%), L4(25%), L5(0%)저농도
    2~3회 반복 측정
  - Y=a+Bx, (x=expected, y= measured) (1차 회귀식-weighted 권장)

- Mechanisms of Interference
  - Chemical effects, Physical effects, Matrix effects, Enzyme inhibition, Nonspecificity, Cross-reactivity, Water displacement
- Interfering Substances
  - metabolites produced in pathological conditions
  - compounds introduced during patient treatment
  - substances ingested
  - substances added during specimen preparation
  - contaminants inadvertently introduced during specimen handling
  - the specimen matrix itself

- Interference testing
  - 허용기준의 설정
  - 현실적인 시험선택
  - 간섭물질에 대해 농도-반응관계를 정량적 제시
  - 간섭물질농도표→EP37
- 제조사의 간섭자료를 기본으로 하여 HIL index를 일상운영

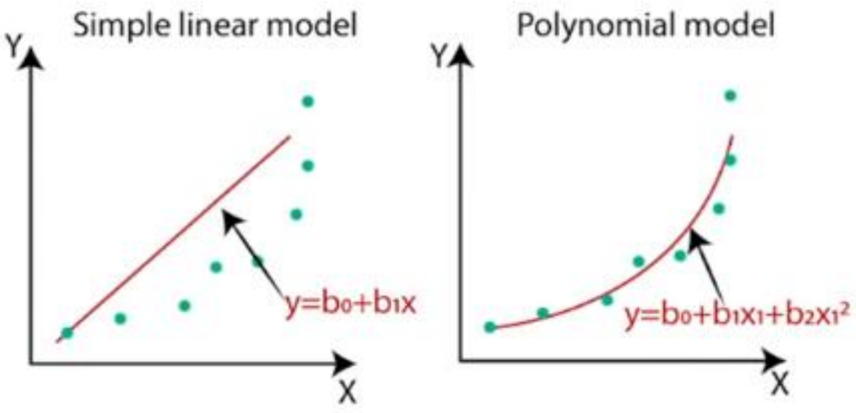
- 환자검체를 이용한 측정법 비교 및 바이어스 추정: 서로 다른 두 검사법 간의 결과를 비교하고 발생하는 오차(바이어스)를 통계적으로 분석
- 정량검사방법간 결과 판정비교를 통한 바이어스 추정
- 다양한 농도 검체를 여러 번 측정 후 이 결과를 회귀분석→ Proportional bias(비례치우침)과 Constant bias(상존치우침)을 추정
  - 의학적 의사결정에 중요한 농도(MDP)에서 차이가 허용가능한 수준인지 판정((Allowable bias)를 기준으로 적용)
- 왜 비교하는가?
  - 두 방법이 같은 값을 내느냐 보다 두 방법사이의 bias가 얼마나 되는가?
  - 실제 환자검체의 matrix, 임상분포를 반영하는 것이 bias평가에 적절
  - Difference plot, scatter plot-시각화

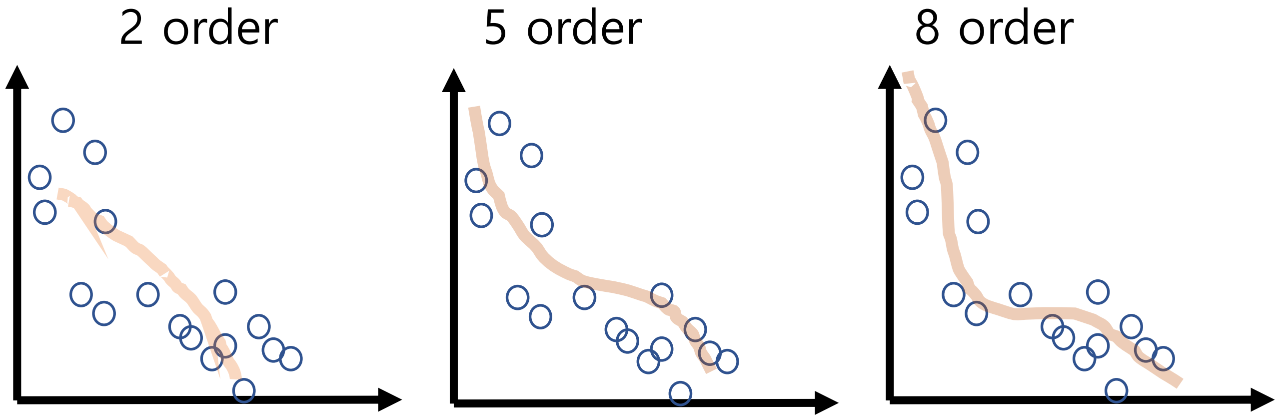
- 생화학- 연속형 결과, 측정범위 전반에서 안정적인 분산구조
  - TEa 기준 적용
- 면역학- EP07 간섭, EP17저농도 성능, EP12정성판정문서 등과 같이 검토 필요
- 혈액학-H 문서를 같이 검토해야 함,
- 상관계수의 함정
  - Bias, slope 같이 고려해야 함
- 기존장비 vs. 새 장비
- Pairwise비교 시 anchor method 기준을 설정
- 시약lot변경 시 동등성 평가 →EP26

Deming regression	Passing-Bablok regression
오차구조확인(CV, Sd있음)	CV, SD모름
비교적 정확	비모수적 접근
반복 측정 자료 있음	Outlier에 강함
연구 논문	데이터가 어떻게 생겼는지 모른다
method validation	precision profile없음
reference method비교	안전함
slope는 오차비율(λ)=$\sigma x^2/\sigma y^2$	slope는 모든 쌍의 기울기의 median
	precision이 반영되지 않는다

Polynomial regression(다항회귀식)

- 모델을 다차원다항식으로 두고 회귀식을 구하는 것



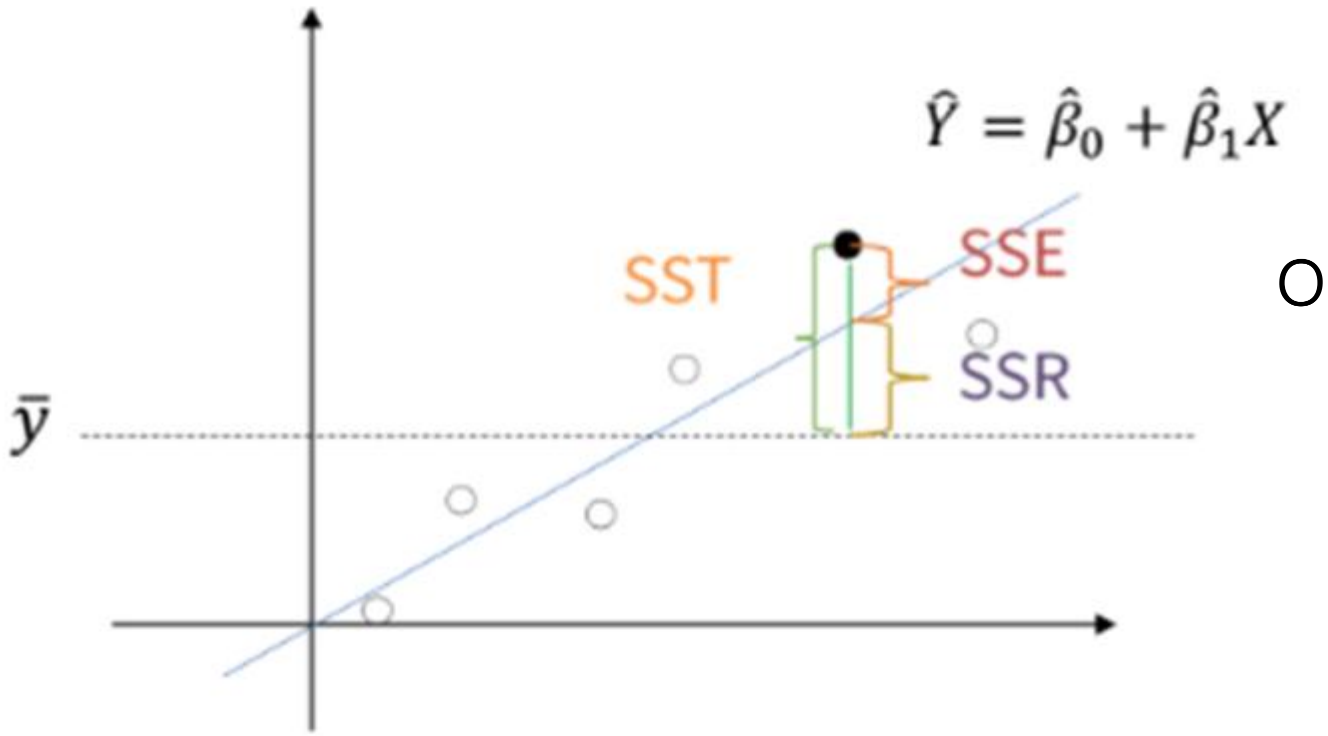


어떤 model이 이 변화를 최적으로 설명하는가?

$$y = \beta_0 + \beta_1 x$$

회귀계수(parameter)

선형 회귀의 정확도 평가



$$y_i - \bar{y} = y_i - \hat{y}_i + \hat{y}_i - \bar{y}$$

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{\substack{\text{SST} \\ \text{(Total sum of squares)}}} = \underbrace{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\substack{\text{SSE} \\ \text{(Error sum of squares)}}} + \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{\substack{\text{SSR} \\ \text{(Regression sum of squares)}}}$$

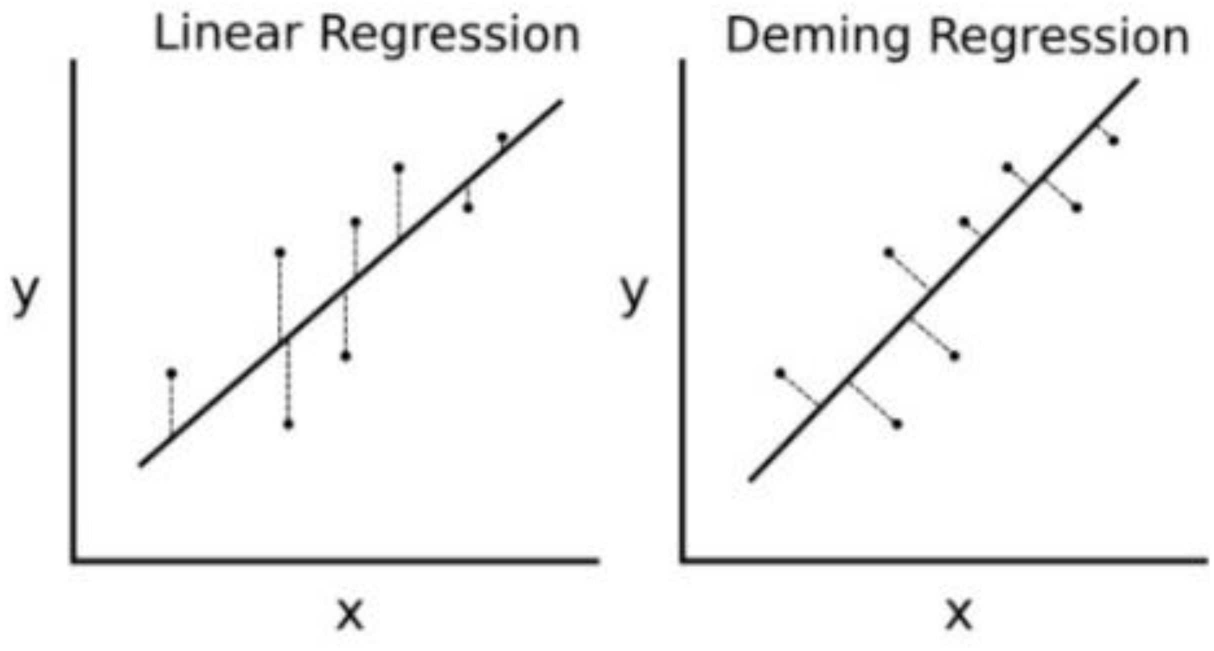
OLS(Ordinary least square)회귀분석

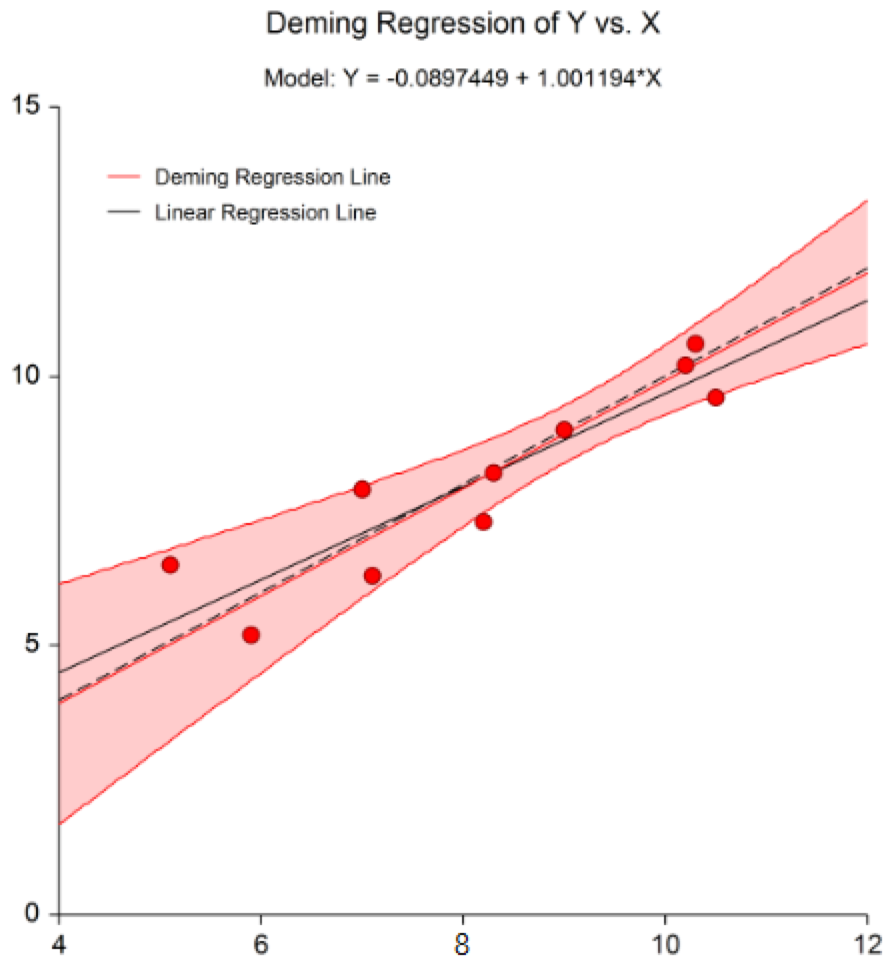
- 최소제곱법을 이용해서 population의 paramete를 추정하여 회귀분석하는 것

Deming Regression(Total regression)

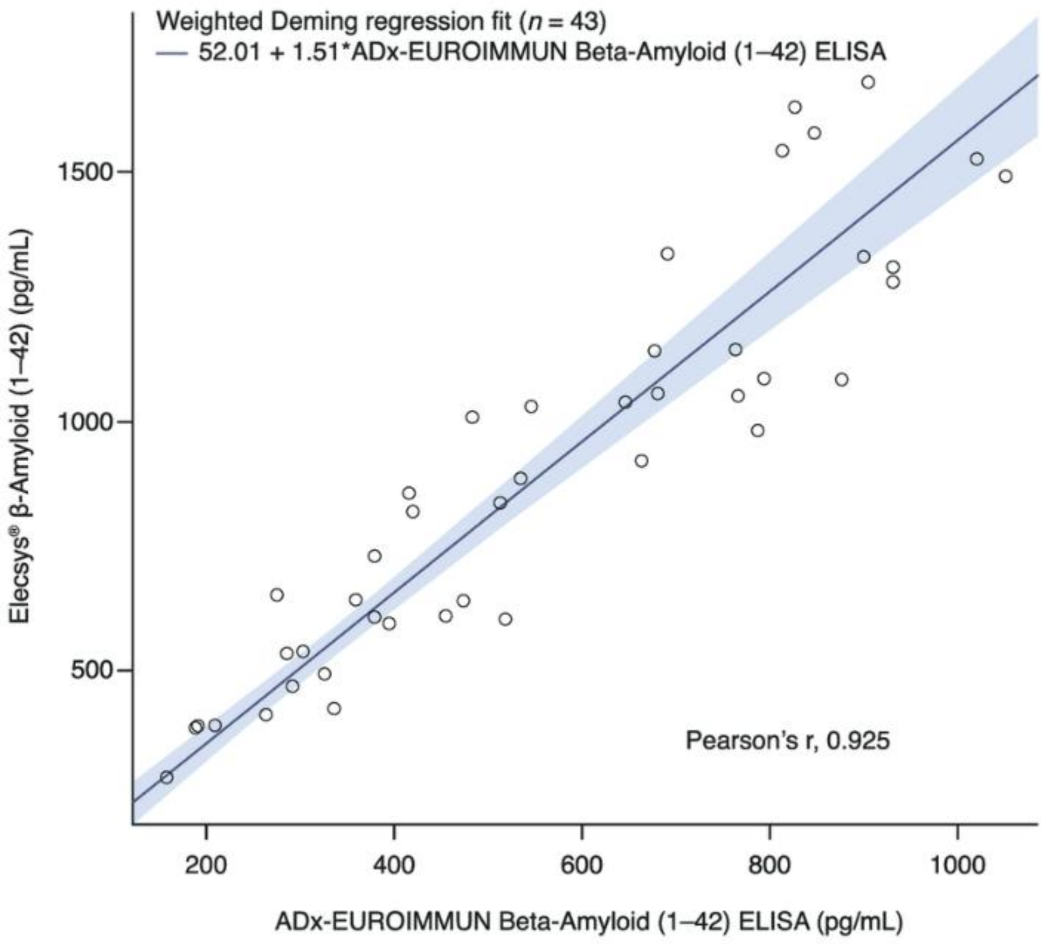
=직교 회귀 분석

- 기존 simple linear regression과 다른 점은 x-, y-축 모두에서 관찰되는 error를 모두 반영하여 계산한 것





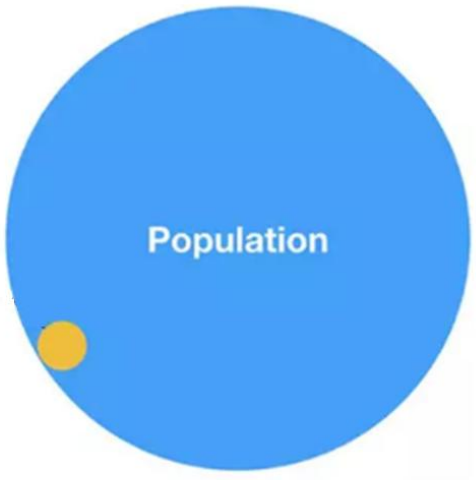
Deming Regression of Y vs. X
Model: Y = -0.0897449 + 1.001194*X
Deming Regression Line
Linear Regression Line
15
10
5
0
4
6
8
10
12

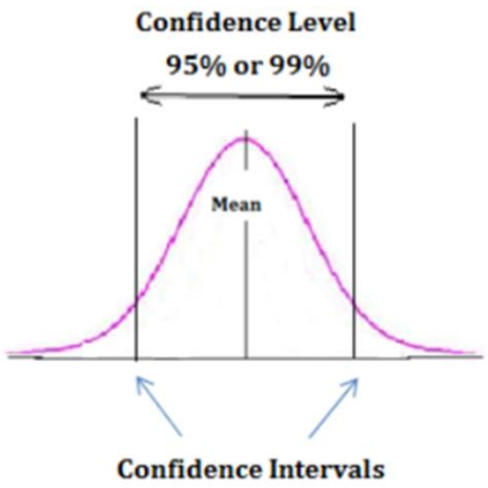


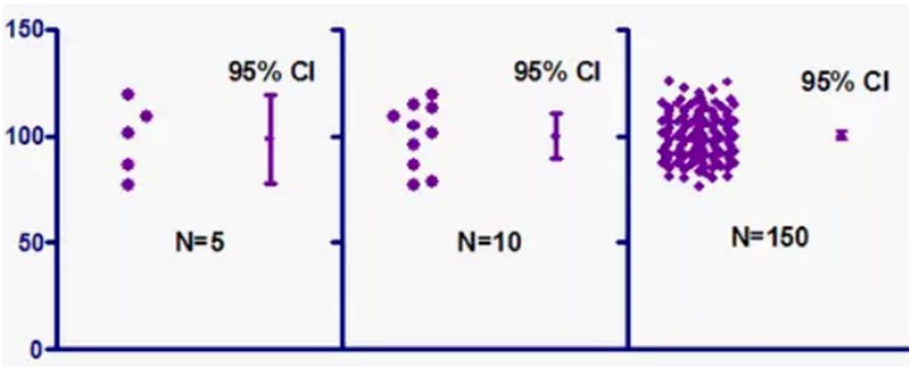
Weighted Deming regression fit (n = 43)
52.01 + 1.51*ADx-EUROIMMUN Beta-Amyloid (1–42) ELISA
Elecsys® β-Amyloid (1–42) (pg/mL)
1500
1000
500
200
400
600
800
1000
Pearson's r, 0.925
ADx-EUROIMMUN Beta-Amyloid (1–42) ELISA (pg/mL)

신뢰구간(confidence interval; CI)

- 신뢰구간은 모수가 실제로 포함될 것으로 예측되는 범위, 즉 샘플링된 표본이 실제 모집단을 얼마나 잘 대표하는지 측정하는 방법
- 신뢰구간에 모집단 실제 평균값이 포함될 확률을 CI의 신뢰수준이라함







- Cut-off 근처
  - 정밀도, 정확도, 선형성, 비교 연구 등 대부분에서 이산형 결과는 cut-off 주위를 확인해야 함
  - Cut-off 에서 bias가 발생하면 직선성이 아무리 좋아도 의미가 없음
  - Agreement가 높은 검사≠좋은 검사, 반드시 borderline sample을 집중해야 함→ cut-off ±20% 구간 샘플 확보하고 discordant분석해야
  - COI, S/CO는 semi–quantitative index, '상대 반응값'
    - 농도값이 아니다-숫자가 아닌 경계에서의 안정성을 검토해야함
    - cut-off 주변 분석 필수
    - Gray zone설정 고려

- 검출능은 분석물질의 존재여부를 검출하는 한계를 정의, 측정가능구간(Analytical measuring interval, AMI)의 하한을 정의
- LOB(공시료 측정치의 한계), LOD(검출 한계), LOQ(정량 한계)
- 검출능의 검정
  - 최종 사용자인 검사실에서 특정 검사법의 수행능 특성이 제조사가 설정한 목표치를 충족시키는 지를 평가
- 실험설계:
  - 1개 제조번호 시약사용, 분석장비 또는 시스템 1대사용, 3일 이상 검증 수행하고 검체 2개(LOB의 경우 공시료, LOD, LOQ의 경우 저농도검체) 이상 사용하며 최소 20회 이상의 결과 확보
  - 검정:
    - LOB검정: 제조사 제시 LOB값보다 같거나 작은 측정값의 개수
    - LOD검정: 제조사 제시 LOB값보다 같거나 큰 측정값의 개수
    - LOQ검정: 총 오차목표 이내에 드는 측정값의 개수

EP19 A Framework for Using CLSI Documents to Evaluate Medical Laboratory Test Methods

- 임상검사실 측정법 평가 체계

구분	검증(validation)	검정(verification)
주체	제조사, 개발자	검사실(최종 사용자)
핵심활동	성능 특성의 설정 및 확정	제조사 목표치(claim)의 이행확인
CLSI 지침 예시	EP05-A3, EP06	EP15-A3,EP34

EP21 Plus Evaluation of Total Analytical Error for Quantitative Medical Laboratory Measurement Procedures

- 총분석오차(total analytical error, TAE)를 추정하고 평가하기 위한 표준화된 방법론
  - $TAE=|Bias|+z*SD_{WL2}$
  - 비모수적 접근-실제 환자 검체 데이터를 기반으로 측정됨
- ATE 설정
  - 임상결과 기반 모델, 생물학적 변이 데이터,동료집단 및 현재 기술수준
  - EP46에 한계설정 내용을 분리

EP24 Assessment of the Diagnostic Accuracy of Laboratory Tests Using Receiver Operating Characteristic Curves

- 임상검사실 측정법 평가 체계
  - ROC 곡선 활용-AUC를 통해 진단성능을 판단
  - Cut-off설정
- 신규검사법 도입 시 임상적 유효성 평가의 도구로 활용됨

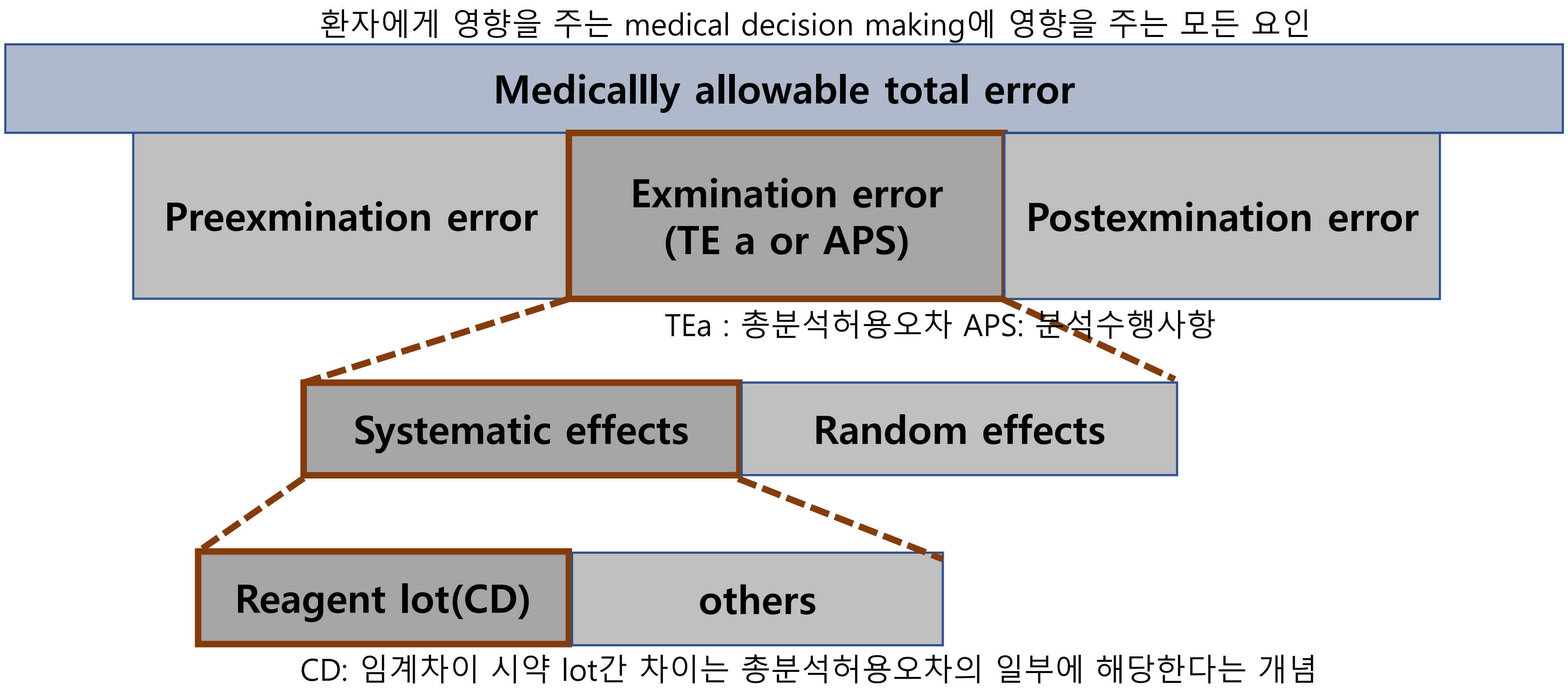
EP25 Evaluation of Stability of In Vitro Medical Laboratory Test Reagents

- 시약, 교정 물질, 정도관리 물질 등이 명시된 기간 내에 분석적 성능을 일관되게 유지하는 지를 확인하여 검사 결과의 신뢰성을 보장하기 위한 지침서

EP26 User Evaluation of Acceptability of a Reagent Lot Change

- 시약 로트 변경 시 동등성 평가-새LOT가 수용가능한가?
- |평균차이|<거절한계: acceptable

TEa 와 CD의 이해


환자에게 영향을 주는 medical decision making에 영향을 주는 모든 요인
Medicallly allowable total error
Preexmination error
Exmination error
(TE a or APS)
Postexmination error
TEa : 총분석허용오차 APS: 분석수행사항
Systematic effects
Random effects
Reagent lot(CD)
others
CD: 임계차이 시약 lot간 차이는 총분석허용오차의 일부에 해당한다는 개념

EP28 Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory

- 건강인 집단을 기준으로 정상범위를 만들고, 각 검사실에서도 그 범위가 맞는지 확인해야함
  - Reference population 정의, reference sample선정, 통계적 계산, 간이 검정
  - 신규검사 도입시, 장비 변경 시, 시약변경(면역), 환자군의 변화, reference range가 의심될 때➔변화가 있을 때
  - Verification
    - 20명 검사, 10%

EP31 PLUS Verification of Comparability of Patient Results Within One Health Care System

- 검사실이 보유한 여러 장비 간의 결과 차이를 정밀하게 측정하고 관리, 환자 진료에 일관된 데이터를 제공할 수 있도록 돕는 실무 통합형 지침(정밀도 추정, 검체선정, 허용기준 설정, 측정횟수, 반복수, 범위기준 설정, 측정, 통계)

면역학

문서	주제	언제 적용하나	기본 설계	대표 계산식/평가지표	실무 해석 포인트	비고
EP05	정밀도 평가	면역분석기 정량검사 CV 확인	저·중·고 농도 반복 측정	CV%=SD/평균×100	저농도, cut-off 인근 CV 중요	cut-off 인근 imprecision이 판정 변동성에 영향
EP07	간섭 평가	이종항체, 류마티스인자, biotin 등 간섭 검토	잠재 간섭물질 첨가 전후 비교	%Interference=(간섭-대조)/대조×100	면역학은 비특이 반응으로 임상적 오판정 가능	정성/정량 모두 false result 관리에 중요
EP09	방법 비교	다른 플랫폼/회사 키트 비교	환자 검체 다수, 음성-양성 전 범위 포함	회귀식, concordance, bias	단순 상관계수만으로 판단하면 안됨	의료결정점 또는 cut-off 주변 bias 확인
EP12	정성 검사 평가	양성/음성 판정형 검사 검증	비교법 또는 상태확정군으로 민감도/특이도 평가	Sensitivity=TP/(TP+FN); Specificity =TN/(TN+FP); PPV, NPV	ROC, cut-off, agreement를 함께 본다	QC는 양성·음성 대조물질, cut-off 근처 약양성 관리
EP15	사용자 검증	신규 면역검사 간이 검증	5일 반복 등 단기 설계	평균, SD, CV%	정식 validation 전 초기 사용 적합성 확인	Levey-Jennings 규칙 설정 전 참고
EP17	검출능 평가	초고감도 분석, low positive 관리	blank/low sample 반복	LoB, LoD, LoQ	보고하한과 재검 기준 설정에 중요	저농도 QC와 reflex test 정책 설계
EP28	참고 구간	정량 면역검사의 참고구간 설정	참고개체 검토	백분위수 기반 구간	면역표지자마다 인구집단 차이 큼	해석 기준 관리

생화학

문서	주제	언제 적용하나	기본 설계	대표 계산식/ 평가지표	실무 해석 포인트	비고
EP05	정밀도 평가	신규 시약/장비 도입, QC 재설정, 성능 확인	보통 2개 이상 농도, 최소 20일 반복 측정 설계	평균=SUM(x)/n; SD=sqrt(sum((xi-x̄)^2)/(n-1)); CV%=SD/평균×100	농도별 CV%가 허용 범위 이내인지 본다. 낮은 농도는 CV%가 커지기 쉬움	CV는 sigma 계산에 직접 사용: Sigma=(TEa%-Bias%)/CV%
EP06	선형성 평가	측정범위 확인, 희석 직선성 검토	고농도-저농도 혼합으로 5개 이상 수준 준비	Bias(level)=측정값-예상값; %Bias=Bias/예상값×100	각 수준의 편차가 허용 범위 내면 선형성 수용	ADL을 TEa X 0.5
EP07	간섭 평가	용혈/황달/지질/약물 간섭 확인	대조군과 간섭군 비교	Interference=평균(간섭군)-평균(대조군); %Interference=차이/대조군×100	임상적으로 의미 있는 변화인지 확인	%Interference가 TEa 를 초과하는지
EP09	방법 비교	구장비-신장비 비교, 외주/사내 방법 비교	환자 검체 다수, 전 범위 포함	Passing-Bablok 또는 Deming; Bias at medical decision point	기울기 CI에 1 포함 여부, 절편 CI에 0 포함 여부검토	MDP에서 예측 bias를 TEa와 비교
EP15	사용자 검증	짧은 기간의 간이 검증	보통 5일×5회 반복 또는 제조사 주장 검증	평균, SD, CV% 계산 후 제조사/허용치 비교	짧고 실용적이지만 정식 정밀도 평가보다 제한적	도입 초기 QC 허용범위 점검에 적합
EP17	검출능 평가	LoB/LoD/LoQ 설정, 저농도 분석 성능 확인	blank와 low-level sample 반복 측정	LoB≈blank 95th percentile; LoD≈LoB + 1.645×SD(low); LoQ=목표 CV 달성 농도	정량한계는 단순 검출보다 임상적 사용성을 반영	저농도 QC와 보고 하한 설정에 활용
EP28	참고구간	기준범위 신규 설정/전이 검증	참고개체 모집 또는 기존 구간 검증		성별/연령/장비차에 따른 재검토 필요	QC보다 결과 해석 체계 정비에 가까움

혈액학

문서	주제	언제 적용하나	기본 설계	대표 계산식/평가지표	실무 해석 포인트	비고
H26	자동혈구분석기 평가	CBC 장비 도입/교체, 성능 비교	반복성, carryover, 정확성, 비교성 설계	Carryover%=(L1-L3)/(H3-L3)×100 등	PLT/WBC/HGB 등 항목별 허용기준 검토	항목별 QC 물질과 moving average 병행 가능
H20	백혈구 감별 평가	WBC differential 비교	수기 diff 또는 비교기기와 다수 검체 비교	agreement%, correlation, bias	monocyte/eosinophil/basophil은 변동성 커 해석 주의	비정상 플래그 검토 규칙과 연결
EP05	정밀도 평가	CBC 재현성 확인	정상/저농도/고농도 반복	CV%=SD/평균×100	PLT 저농도, retic 등은 별도 주의	Westgard 규칙보다 moving average/델타체크 병행 유용
EP09	방법 비교	장비 간 CBC 비교	환자 검체 다수, 저·고 범위 포함	회귀식, bias, Bland-Altman	WBC diff는 상관보다 개별 세포군 bias를 중점 검토	허용편차와 임상 영향 함께 판단
EP15	사용자 검증	단기 도입 검증	5일 반복 등	평균, SD, CV%	현장 적용 전 신속 확인	초기 QC 범위/재측정 기준 참고